Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. Collage of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding, and benchmarked task-specific models including RetinaFace, DeepFace, and Siamese architectures for targeted detection, face analysis, and visual tracking/re-identification.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Worked on multimodal perception for autonomous systems by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJECTS

Camera-Based Perception and Simulation for Autonomous Vehicles (TIHAN, IIT Hyderabad):

- Developed vision-based perception systems using YOLO and DeepSORT for real-time object detection, multi-object tracking, and position estimation of surrounding vehicles and pedestrians.
- Tested autonomous vehicle perception and behavior in CARLA simulation across driving scenarios before real-world testing.
- Implemented perception modules with ROS2 on Linux for vehicle integration and testing.
- Performed LiDAR-based and camera data annotation for training and validating perception models.

Multimodal Vision-Language Model Evaluation for Computer Vision Tasks (LFOVIA Lab, IIT Hyderabad):

- Evaluated generative-AI Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision) for scene understanding, including object detection, segmentation, crowd analysis, action recognition, and text recognition, as alternatives to task-specific models.
- Explored VLM-based generative AI for scene understanding in autonomous systems, interpreting road users, signals, and scene context from camera images.
- Integrated model inference with ROS2 on Linux to support real-time perception pipelines.
- Benchmarked and fine-tuned models using Python-based pipelines, evaluating detection performance (precision/recall, IoU).

Multi-Channel Conversational AI Platform (IIT Hyderabad Build Program):

- Built a multi-tenant generative-AI platform unifying voice, chat, and email, with a hybrid retrieval pipeline (semantic + keyword) and LLM-based intent classification that improved answer relevance by 18%.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and experimented with Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict hydrogen production from reactor operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV).
- Recently adapted the RRSI method to this work and experimenting with it to rigorously verify that model improvements.

TECHNICAL SKILLS

Programming & Tools: Python, C++, SQL, Linux, ROS2, OpenCV, PyTorch, TensorFlow, CUDA, NumPy, Pandas, Scikit-learn

Autonomous Systems & Computer Vision: CARLA, Gazebo, Camera-Based Perception, Object Detection & Multi-Object Tracking (YOLO, DeepSORT), Position Estimation, Face Analysis (DeepFace, RetinaFace), OCR, Data Annotation (LiDAR, Camera, NIR, Sonar)

Generative AI & Multimodal: Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), Large Language Models (LLMs), LLM Fine-Tuning, RAG, Transformer Architectures, CNN

Model Evaluation: Precision, Recall, IoU, Model Benchmarking

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs).” IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging.” IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS & ACHIEVEMENTS

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant